

Week 08

1 Bayes Classifier

Bayes classifier

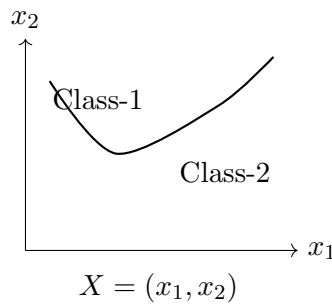
$$\delta(x) = \underset{1 \leq j \leq J}{\operatorname{argmax}} \pi_j f_j(x) = \underset{1 \leq j \leq J}{\operatorname{argmax}} p(j | x).$$

Class-1 $N(\mu_1, \Sigma_1)$: π_1 Class-2 $N(\mu_2, \Sigma_2)$: π_2 .

We classify an obs. x to Class-1 if $\pi_1 f_1(x) > \pi_2 f_2(x)$

$$\begin{aligned} &\Leftrightarrow \pi_1 \frac{1}{(2\pi)^{d/2} |\Sigma_1|^{1/2}} e^{-\frac{1}{2}(x-\mu_1)'\Sigma_1^{-1}(x-\mu_1)} > \pi_2 \frac{1}{(2\pi)^{d/2} |\Sigma_2|^{1/2}} e^{-\frac{1}{2}(x-\mu_2)'\Sigma_2^{-1}(x-\mu_2)} \\ &\Leftrightarrow \log \pi_1 - \frac{1}{2} \log |\Sigma_1| - \frac{1}{2}(x-\mu_1)'\Sigma_1^{-1}(x-\mu_1) \\ &\quad > \log \pi_2 - \frac{1}{2} \log |\Sigma_2| - \frac{1}{2}(x-\mu_2)'\Sigma_2^{-1}(x-\mu_2) \\ &\Leftrightarrow \frac{1}{2}(x-\mu_2)'\Sigma_2^{-1}(x-\mu_2) - \frac{1}{2}(x-\mu_1)'\Sigma_1^{-1}(x-\mu_1) > \log\left(\frac{\pi_2 |\Sigma_1|^{1/2}}{|\Sigma_2|^{1/2} \pi_1}\right). \end{aligned}$$

This is a quadratic classifier.



The above inequality can be simplified as

$$\frac{1}{2}(x-\mu_2)'\Sigma^{-1}(x-\mu_2) - \frac{1}{2}(x-\mu_1)'\Sigma^{-1}(x-\mu_1) > \log(\pi_2/\pi_1)$$

2 Equal Covariance Case – Linear Classifier

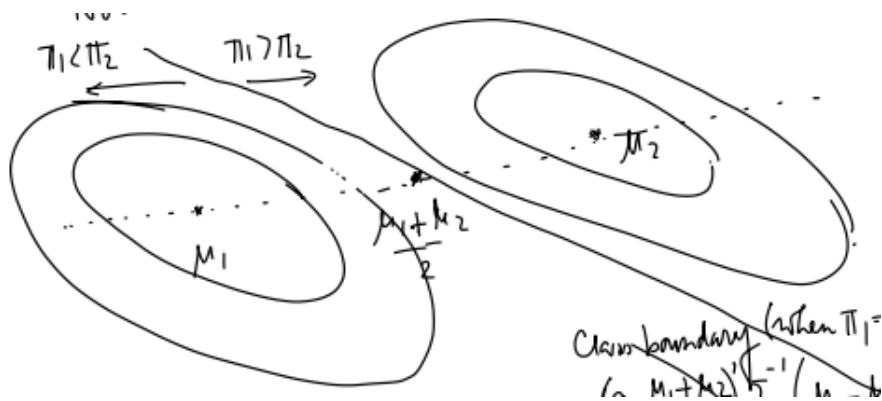
Expanding:

$$\begin{aligned} &\{x'\Sigma^{-1}x - \mu_2'\Sigma^{-1}x - x'\Sigma^{-1}\mu_2 + \mu_2'\Sigma^{-1}\mu_2\} \\ &- \{x'\Sigma^{-1}x - \mu_1'\Sigma^{-1}x - x'\Sigma^{-1}\mu_1 + \mu_1'\Sigma^{-1}\mu_1\} > 2\log(\pi_2/\pi_1) \\ &\Leftrightarrow 2x'\Sigma^{-1}(\mu_1 - \mu_2) + \mu_2'\Sigma^{-1}\mu_2 - \mu_1'\Sigma^{-1}\mu_1 > 2\log(\pi_2/\pi_1) \\ &\Leftrightarrow x'\Sigma^{-1}(\mu_1 - \mu_2) + \frac{1}{2}[\mu_2'\Sigma^{-1}\mu_2 - \mu_1'\Sigma^{-1}\mu_1] > \log(\pi_2/\pi_1) \\ &\Leftrightarrow \left(x - \frac{\mu_1 + \mu_2}{2}\right)'\Sigma^{-1}(\mu_1 - \mu_2) > \log(\pi_2/\pi_1). \end{aligned}$$

Note that this is a linear classifier.

$$\left(\frac{\mu_1 + \mu_2}{2} \right)' \Sigma^{-1} (\mu_1 - \mu_2)$$

$$\frac{1}{2} \left[\mu_1' \Sigma^{-1} \mu_1 - \cancel{\mu_1' \Sigma^{-1} \mu_2} + \cancel{\mu_2' \Sigma^{-1} \mu_1} - \mu_2' \Sigma^{-1} \mu_2 \right]$$



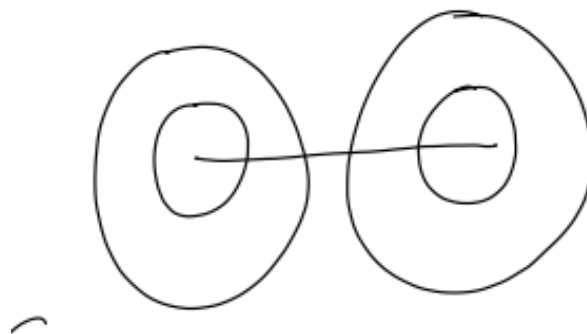
Class boundary (when $\pi_1 = \pi_2$):

$$\left(x - \frac{\mu_1 + \mu_2}{2} \right)' \Sigma^{-1} (\mu_1 - \mu_2) = 0.$$

If $\pi_1 = \pi_2 = 1/2$ then we have

$$\left(x - \frac{\mu_1 + \mu_2}{2} \right)' \Sigma^{-1} (\mu_1 - \mu_2) > 0.$$

If $\Sigma = I$, then the separating surface will be $\perp$ to the line joining μ_1 and μ_2 ($\perp$ to $\mu_1 - \mu_2$).



3 Misclassification Probability of the Linear Classifier

Classifier δ_ℓ

$$\delta_\ell(x) = 1 \quad \text{if} \quad \left(x - \frac{\mu_1 + \mu_2}{2}\right)' \Sigma^{-1}(\mu_1 - \mu_2) > \log(\pi_2/\pi_1).$$

Misclassification prob. of δ_ℓ :

$$\begin{aligned} &= \pi_1 P\left[\left(x - \frac{\mu_1 + \mu_2}{2}\right)' \Sigma^{-1}(\mu_1 - \mu_2) < \log(\pi_2/\pi_1) \mid X \sim N(\mu_1, \Sigma)\right] \\ &+ \pi_2 P\left[\left(x - \frac{\mu_1 + \mu_2}{2}\right)' \Sigma^{-1}(\mu_1 - \mu_2) > \log(\pi_2/\pi_1) \mid X \sim N(\mu_2, \Sigma)\right]. \end{aligned}$$

$$X \sim N(\mu_1, \Sigma) \implies \ell'X \sim N(\ell'\mu_1, \ell'\Sigma\ell), \quad \ell'X + a \sim N(\ell'\mu_1 + a, \ell'\Sigma\ell).$$

Let $\ell = \Sigma^{-1}(\mu_1 - \mu_2)$, $a = -\frac{1}{2}(\mu_1 + \mu_2)' \Sigma^{-1}(\mu_1 - \mu_2)$:

$$\begin{aligned} \left(x - \frac{\mu_1 + \mu_2}{2}\right)' \Sigma^{-1}(\mu_1 - \mu_2) &= \underbrace{x' \Sigma^{-1}(\mu_1 - \mu_2)}_{\ell} - \underbrace{\frac{1}{2}(\mu_1 + \mu_2)' \Sigma^{-1}(\mu_1 - \mu_2)}_a \\ &\sim N\left[\frac{1}{2}(\mu_1 - \mu_2)' \Sigma^{-1}(\mu_1 - \mu_2), (\mu_1 - \mu_2)' \Sigma^{-1} \Sigma \Sigma^{-1}(\mu_1 - \mu_2)\right] \equiv N\left[\frac{1}{2}\Delta^2, \Delta^2\right] \end{aligned}$$

where Δ is the Mahalanobis distance.

Similarly, if $X \sim N(\mu_2, \Sigma)$,

$$\left(x - \frac{\mu_1 + \mu_2}{2}\right)' \Sigma^{-1}(\mu_1 - \mu_2) \sim N\left[-\frac{1}{2}\Delta^2, \Delta^2\right].$$

Misclassification probability

$$\begin{aligned} &= \pi_1 P\left[\left(x - \frac{\mu_1 + \mu_2}{2}\right)' \Sigma^{-1}(\mu_1 - \mu_2) < \log(\pi_2/\pi_1) \mid X \sim N(\mu_1, \Sigma)\right] \\ &+ \pi_2 P\left[\left(x - \frac{\mu_1 + \mu_2}{2}\right)' \Sigma^{-1}(\mu_1 - \mu_2) > \log(\pi_2/\pi_1) \mid X \sim N(\mu_2, \Sigma)\right]. \\ &= \pi_1 P\left[N\left(\frac{1}{2}\Delta^2, \Delta^2\right) < \log(\pi_2/\pi_1)\right] + \pi_2 P\left[N\left(-\frac{1}{2}\Delta^2, \Delta^2\right) > \log(\pi_2/\pi_1)\right] \\ &= \pi_1 P\left[Z < \frac{\log(\pi_2/\pi_1) - \frac{1}{2}\Delta^2}{\Delta}\right] + \pi_2 P\left[Z > \frac{\log(\pi_2/\pi_1) + \frac{1}{2}\Delta^2}{\Delta}\right] \\ &= \pi_1 \cdot \Phi\left(\frac{\log(\pi_2/\pi_1) - \frac{1}{2}\Delta^2}{\Delta}\right) + \pi_2 \left\{1 - \Phi\left(\frac{\log(\pi_2/\pi_1) + \frac{1}{2}\Delta^2}{\Delta}\right)\right\}. \end{aligned}$$

If $\pi_1 = \pi_2$, then this prob is

$$= \frac{1}{2}\Phi\left(-\frac{\Delta}{2}\right) + \frac{1}{2}\left[1 - \Phi\left(\frac{\Delta}{2}\right)\right] = \Phi\left(-\frac{\Delta}{2}\right) = 1 - \Phi\left(\frac{\Delta}{2}\right).$$

So, this probability is a decreasing function of the Mahalanobis distance.

4 Estimation from Data

In practice, we do not know the density functions or posterior probabilities of the competing classes. We need to estimate from the data.

$$\begin{array}{ll} \text{Class-1: } \pi_1 & | \quad X_{11}, \dots, X_{1n_1} \\ \text{Class-2: } \pi_2 & | \quad X_{21}, \dots, X_{2n_2} \end{array}$$

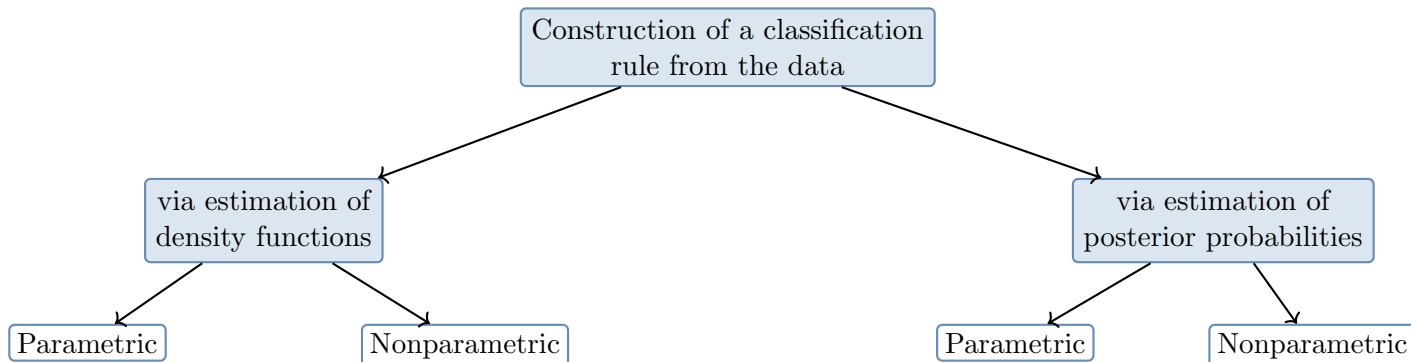
These observations are to be used to estimate $\hat{f}_1$ & $\hat{f}_2$ or $\hat{p}(1 | x)$ & $\hat{p}(2 | x)$.

In such cases, we plug-in these estimates to get the classifier of the form

$$\delta(x) = \operatorname{argmax}_{1 \leq j \leq J} \pi_j \hat{f}_j(x) \quad (\text{also estimate the } \pi_j \text{'s if they are not known})$$

$$= \operatorname{argmax}_{1 \leq j \leq J} \hat{p}(j | x), \quad \hat{p}(1 | x) = \frac{\pi_1 \hat{f}_1}{\pi_1 \hat{f}_1 + \pi_2 \hat{f}_2}.$$

5 Construction of a Classification Rule from Data



Density estimation route

- **Parametric**

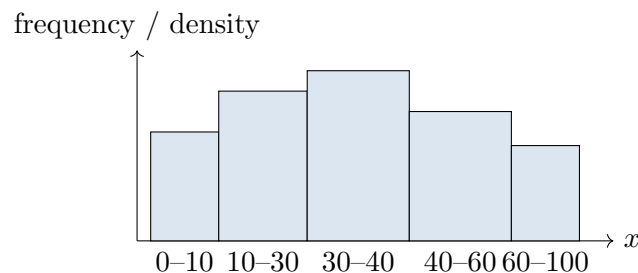
We assume a parametric form for the underlying densities and estimate the model parameters.

- LDA (linear discriminant analysis)
- QDA (quadratic discriminant analysis)

- **Nonparametric**

We estimate the densities nonparametrically.

- KDA (kernel density analysis)
- Classifier based on nearest-neighbour density estimates



$$\hat{f}_{\text{Hist}}(r) = \frac{r_i}{k_i \cdot n} = \frac{\text{frequency}}{\text{width} * \text{number of observations}}$$

Interval	freq	freq density
0-10	20	2
10-30	30	1.5
30-40	40	4
40-60	50	2.5
60-100	50	1.25

Posterior probability route

- **Parametric**

We assume a parametric form for the posterior probabilities of different classes and estimate the model parameters from the data.

- Logistic regression

$$\log \frac{p(1 | x)}{1 - p(1 | x)} = \alpha^\top x + \beta,$$

$$p(1 | x) = \frac{e^{\alpha^\top x + \beta}}{1 + e^{\alpha^\top x + \beta}},$$

$$p(2 | x) = \frac{1}{1 + e^{\alpha^\top x + \beta}}.$$

- **Nonparametric** We assume the posterior probabilities non parametrically.

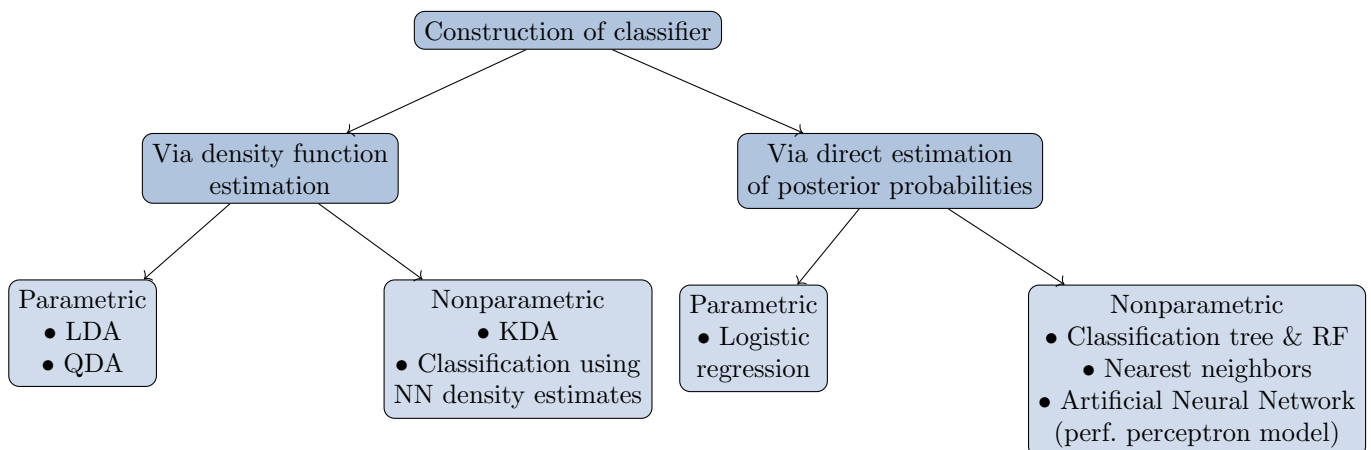
- Classification tree
- Random forest
- Nearest-neighbour classification
- Neural networks (perceptron models)

Support Vector Machine

is another popular classifier which adopts a different approach:

It usually assumes a parametric form for the separating surface and uses a non-parametric method for optimising a suitable objective function.

5.1 Construction of a Classifier



Support vector machine does not belong to any of these above classes. There, we adopt a completely different approach.

6 LDA and QDA

Here we use a parametric method for density estimation. We assume that the underlying distributions are normal and estimate the parameters of the normal distributions from the data.

Class-1: $X_{11}, X_{12}, \dots, X_{1n_1}$

Class-2: $X_{21}, X_{22}, \dots, X_{2n_2}$

with priors $\pi_1 f_1 \equiv \pi_2 f_2$.

We assume that f_1 is the density of $N(\mu_1, \Sigma_1)$ and f_2 is the density of $N(\mu_2, \Sigma_2)$ for some $\mu_1, \mu_2, \Sigma_1, \Sigma_2$.

Usual estimates for $\mu_1, \mu_2, \Sigma_1, \Sigma_2$

$$\begin{aligned}\hat{\mu}_1 &= \bar{X}_1 = \frac{1}{n_1} \sum_{j=1}^{n_1} X_{1j} & \hat{\Sigma}_1 &= \frac{1}{n_1} \sum_{i=1}^{n_1} (X_{1i} - \bar{X}_1)(X_{1i} - \bar{X}_1)' \\ \hat{\mu}_2 &= \bar{X}_2 = \frac{1}{n_2} \sum_{j=1}^{n_2} X_{2j} & \hat{\Sigma}_2 &= \frac{1}{n_2} \sum_{i=1}^{n_2} (X_{2i} - \bar{X}_2)(X_{2i} - \bar{X}_2)'\end{aligned}$$

We classify an obs. x to class-1 if

$$\pi_1 \hat{f}_1(x) > \pi_2 \hat{f}_2(x)$$

where

$$\begin{aligned}\hat{f}_1(x) &= \frac{1}{(2\pi)^{d/2} |\hat{\Sigma}_1|^{1/2}} e^{-\frac{1}{2}(x - \hat{\mu}_1)' \hat{\Sigma}_1^{-1} (x - \hat{\mu}_1)}, & \hat{f}_2(x) &= \frac{1}{(2\pi)^{d/2} |\hat{\Sigma}_2|^{1/2}} e^{-\frac{1}{2}(x - \hat{\mu}_2)' \hat{\Sigma}_2^{-1} (x - \hat{\mu}_2)}. \\ \pi_1 \frac{1}{(2\pi)^{d/2} |\hat{\Sigma}_1|^{1/2}} e^{-\frac{1}{2}(x - \hat{\mu}_1)' \hat{\Sigma}_1^{-1} (x - \hat{\mu}_1)} &> \pi_2 \frac{1}{(2\pi)^{d/2} |\hat{\Sigma}_2|^{1/2}} e^{-\frac{1}{2}(x - \hat{\mu}_2)' \hat{\Sigma}_2^{-1} (x - \hat{\mu}_2)} \\ &\Leftrightarrow \log\left(\pi_1 |\hat{\Sigma}_1|^{-1/2}\right) - \frac{1}{2}(x - \hat{\mu}_1)' \hat{\Sigma}_1^{-1} (x - \hat{\mu}_1) \\ &> \log\left(\pi_2 |\hat{\Sigma}_2|^{-1/2}\right) - \frac{1}{2}(x - \hat{\mu}_2)' \hat{\Sigma}_2^{-1} (x - \hat{\mu}_2) \\ &\Leftrightarrow \frac{1}{2} \left[(x - \hat{\mu}_2)' \hat{\Sigma}_2^{-1} (x - \hat{\mu}_2) - (x - \hat{\mu}_1)' \hat{\Sigma}_1^{-1} (x - \hat{\mu}_1) \right] > \log\left(\frac{\pi_2 |\hat{\Sigma}_1|^{1/2}}{\pi_1 |\hat{\Sigma}_2|^{1/2}}\right).\end{aligned}$$

This is a quadratic classifier and this classification method is known as **quadratic discriminant analysis (QDA)**.

If we make an additional assumption and assume that $\Sigma_1 = \Sigma_2$, i.e. f_1 is the density of $N(\mu_1, \Sigma)$ and f_2 is the density of $N(\mu_2, \Sigma)$ for some μ_1, μ_2 and Σ , we get a slightly different type of classifier.

Usual estimates for μ_1, μ_2 and Σ

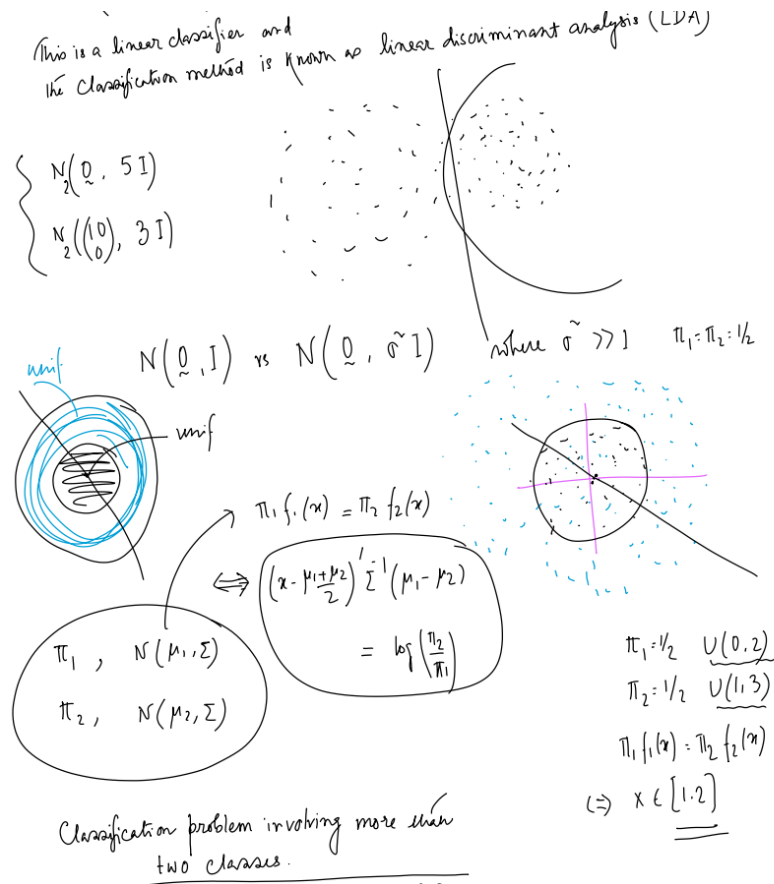
$$\begin{aligned}\hat{\mu}_1 &= \bar{X}_1 = \frac{1}{n_1} \sum_{j=1}^{n_1} X_{1j} & \hat{\mu}_2 &= \bar{X}_2 = \frac{1}{n_2} \sum_{j=1}^{n_2} X_{2j} \\ \hat{\Sigma} &= \frac{\sum_{j=1}^{n_1} (X_{1j} - \bar{X}_1)(X_{1j} - \bar{X}_1)' + \sum_{j=1}^{n_2} (X_{2j} - \bar{X}_2)(X_{2j} - \bar{X}_2)'}{n_1 + n_2}.\end{aligned}$$

We can use these estimates to construct a classifier that classifies an obs. x to class-1 if and only if

$$\pi_1 \frac{1}{(2\pi)^{d/2} |\hat{\Sigma}|^{1/2}} e^{-\frac{1}{2}(x - \hat{\mu}_1)' \hat{\Sigma}^{-1} (x - \hat{\mu}_1)} > \pi_2 \frac{1}{(2\pi)^{d/2} |\hat{\Sigma}|^{1/2}} e^{-\frac{1}{2}(x - \hat{\mu}_2)' \hat{\Sigma}^{-1} (x - \hat{\mu}_2)}$$

$$\begin{aligned}
&\Leftrightarrow \log \pi_1 - \frac{1}{2}(x - \hat{\mu}_1)' \hat{\Sigma}^{-1}(x - \hat{\mu}_1) > \log \pi_2 - \frac{1}{2}(x - \hat{\mu}_1)' \hat{\Sigma}^{-1}(x - \hat{\mu}_2) \\
&\Leftrightarrow \frac{1}{2} \left[(x - \hat{\mu}_2)' \hat{\Sigma}^{-1}(x - \hat{\mu}_2) - (x - \hat{\mu}_1)' \hat{\Sigma}^{-1}(x - \hat{\mu}_1) \right] > \log \left(\frac{\pi_2}{\pi_1} \right) \\
&\Leftrightarrow \frac{1}{2} \left[-2x' \hat{\Sigma}^{-1} \hat{\mu}_2 + \hat{\mu}_2' \hat{\Sigma}^{-1} \hat{\mu}_2 + 2x' \hat{\Sigma}^{-1} \hat{\mu}_1 - \hat{\mu}_1' \hat{\Sigma}^{-1} \hat{\mu}_1 \right] > \log \left(\frac{\pi_2}{\pi_1} \right) \\
&\Leftrightarrow \left(x - \frac{\hat{\mu}_1 + \hat{\mu}_2}{2} \right)' \hat{\Sigma}^{-1} (\hat{\mu}_1 - \hat{\mu}_2) > \log \left(\frac{\pi_2}{\pi_1} \right).
\end{aligned}$$

This is a linear classifier and the classification method is known as **linear discriminant analysis (LDA)**.



7 Classification involving More Than Two Classes

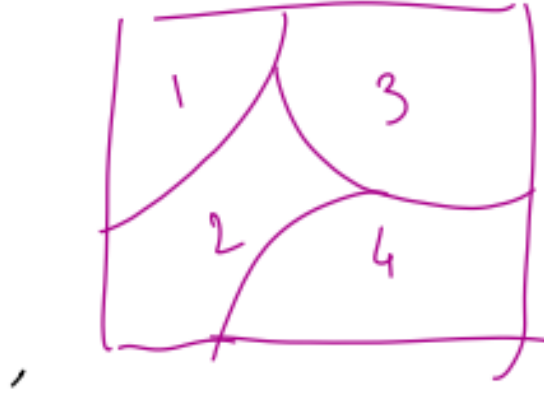
Classification problem involving more than two classes.

QDA

$$\begin{aligned}
\text{Class-1: } X_{11}, X_{12}, \dots, X_{1n_1} \quad \hat{\mu}_1 = \bar{X}_1, \quad \hat{\Sigma}_1 &= \frac{1}{n_1} \sum_{i=1}^{n_1} (X_{1i} - \bar{X}_1)(X_{1i} - \bar{X}_1)' \\
\text{Class-2: } X_{21}, X_{22}, \dots, X_{2n_2} \quad \hat{\mu}_2 = \bar{X}_2, \quad \hat{\Sigma}_2 &= \frac{1}{n_2} \sum_{i=1}^{n_2} (X_{2i} - \bar{X}_2)(X_{2i} - \bar{X}_2)' \\
&\vdots \\
\text{Class-J: } X_{J1}, X_{J2}, \dots, X_{Jn_J} \quad \hat{\mu}_J = \bar{X}_J, \quad \hat{\Sigma}_J &= \frac{1}{n_J} \sum_{i=1}^{n_J} (X_{Ji} - \bar{X}_J)(X_{Ji} - \bar{X}_J)' \\
n &= n_1 + n_2 + \dots + n_J.
\end{aligned}$$

QDA classifier

$$\begin{aligned}
\delta_{\text{QDA}}(x) &= \operatorname{argmax}_{1 \leq j \leq J} \pi_j \hat{f}_j(x) \\
&= \operatorname{argmax}_{1 \leq j \leq J} \pi_j \frac{1}{(2\pi)^{d/2} |\hat{\Sigma}_j|^{1/2}} e^{-\frac{1}{2}(x - \hat{\mu}_j)' \hat{\Sigma}_j^{-1} (x - \hat{\mu}_j)} \\
&= \operatorname{argmax}_{1 \leq j \leq J} \log(\pi_j |\hat{\Sigma}_j|^{-1/2}) - \frac{1}{2}(x - \hat{\mu}_j)' \hat{\Sigma}_j^{-1} (x - \hat{\mu}_j).
\end{aligned}$$



7.1 LDA (More Than Two Classes)

Here we assume $\Sigma_1 = \Sigma_2 = \dots = \Sigma_J$.

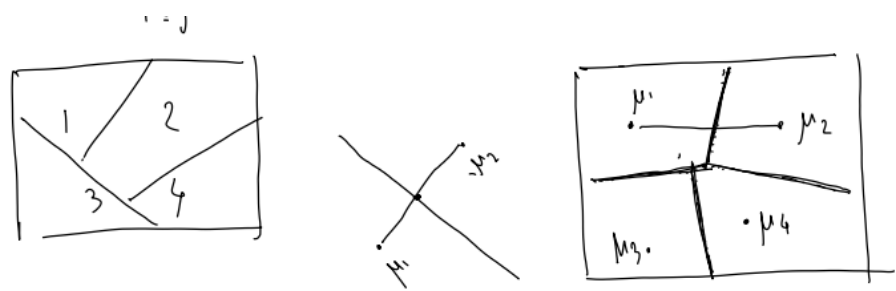
$$\begin{aligned}
\hat{\mu}_1 &= \bar{X}_1, \quad \hat{\mu}_2 = \bar{X}_2, \quad \dots, \quad \hat{\mu}_J = \bar{X}_J \\
\hat{\Sigma} &= \frac{\sum_{j=1}^J \sum_{i=1}^{n_j} (X_{ji} - \bar{X}_j)(X_{ji} - \bar{X}_j)'}{n}
\end{aligned}$$

Replace n by $n - J$ if you look for an unbiased estimator of Σ .

$$\begin{aligned}
\delta_{\text{LDA}}(x) &= \operatorname{argmax}_{1 \leq j \leq J} \pi_j \frac{1}{(2\pi)^{d/2} |\hat{\Sigma}|^{1/2}} e^{-\frac{1}{2}(x - \hat{\mu}_j)' \hat{\Sigma}^{-1} (x - \hat{\mu}_j)} \\
&= \operatorname{argmax}_{1 \leq j \leq J} \pi_j e^{-\frac{1}{2}(x - \hat{\mu}_j)' \hat{\Sigma}^{-1} (x - \hat{\mu}_j)}
\end{aligned}$$

$$= \operatorname{argmax}_{1 \leq j \leq J} \log \pi_j - \frac{1}{2} \left\{ (x - \hat{\mu}_j)' \hat{\Sigma}^{-1} (x - \hat{\mu}_j) \right\}$$

$$\begin{aligned}
&= \operatorname{argmax}_{1 \leq j \leq J} \log \pi_j - \frac{1}{2} \left\{ x' \hat{\Sigma}^{-1} x - 2x' \hat{\Sigma}^{-1} \hat{\mu}_j + \hat{\mu}_j' \hat{\Sigma}^{-1} \hat{\mu}_j \right\} \\
&= \operatorname{argmax}_{1 \leq j \leq J} \log \pi_j - \frac{1}{2} \left[\hat{\mu}_j' \hat{\Sigma}^{-1} \hat{\mu}_j - 2x' \hat{\Sigma}^{-1} \hat{\mu}_j \right].
\end{aligned}$$



$d = 20$		QDA	
Class-1	25	$N(\mu_1, \Sigma)$	$\hat{\Sigma}_1$
Class-2	25	$\vdots$	$\hat{\Sigma}_2$
$\vdots$	$\vdots$		$\vdots$
Class-10	25	$N(\mu_{10}, \Sigma)$	$\hat{\Sigma}_{10}$